

Problem Set 4 – Function families

Session 4 · Polynomials, exponentials, logarithms, and trigonometric models

Part A · Mechanics

1 POLYNOMIAL SHAPE

Let $p(x) = -x^2 + 2x + 3$.

- State the degree and leading coefficient of p .
- Describe what the graph does at its left and right ends.
- Sketch the graph. Give the intercepts with both axes and the vertex, and say how you found each one.

SOLUTION (a) The degree is **2** and the leading coefficient is -1 .

(b) The degree is even and the leading coefficient is negative, so the graph goes down at both ends: $p(x) \rightarrow -\infty$ as $x \rightarrow -\infty$ and as $x \rightarrow \infty$.

(c) *x*-intercepts: factor. $-x^2 + 2x + 3 = -(x^2 - 2x - 3) = -(x + 1)(x - 3)$, so $p(x) = 0$ exactly when $x = -1$ or $x = 3$, giving $(-1, 0)$ and $(3, 0)$.

y-intercept: set $x = 0$, so $p(0) = 3$ and the point is $(0, 3)$.

Vertex: complete the square, $-x^2 + 2x + 3 = -(x - 1)^2 + 4$. Since $(x - 1)^2 \geq 0$ with equality only at $x = 1$, the maximum is **4** at $x = 1$: the vertex is $(1, 4)$. (Equivalently $x = -\frac{b}{2a} = -\frac{2}{-2} = 1$, then $p(1) = 4$.)

The sketch is a downward parabola through $(-1, 0)$, $(0, 3)$, $(3, 0)$, peaking at $(1, 4)$.

2 QUADRATIC TRANSFORMATION

Start with $f(x) = x^2$. The function

$$g(x) = -(x - 2)^2 + 4$$

is made from f .

- Describe the shift and reflection.
- Give the vertex and find the real roots of g .

SOLUTION Move the graph of $y = x^2$ right by **2**, reflect it in the x -axis, then move it up by **4**. Its vertex is $(2, 4)$. For roots, $-(x - 2)^2 + 4 = 0 \iff (x - 2)^2 = 4 \iff x - 2 = \pm 2$, so $x = 0$ or $x = 4$.

3 EXPONENTIAL EQUATION

Solve

$$5^{2x} - 30 \cdot 5^x + 125 = 0.$$

Give all real solutions and justify each step.

SOLUTION Put $u = 5^x$. Then $5^{2x} = (5^x)^2 = u^2$, and the equation becomes the quadratic

$$u^2 - 30u + 125 = 0.$$

The substitution is legitimate because 5^x is defined for every real x , and it takes **only positive values** – which matters when we come back at the end.

Factoring, $u^2 - 30u + 125 = (u - 25)(u - 5)$, so $u = 25$ or $u = 5$. Both are positive, so each gives a genuine solution.

Returning to x : since 5^t is one-to-one (the base 5 is positive and not 1),

$$5^x = 25 = 5^2 \implies x = 2, \quad 5^x = 5 = 5^1 \implies x = 1.$$

The real solutions are $x = 2$ and $x = 1$.

Check. At $x = 2$: $625 - 750 + 125 = 0$. At $x = 1$: $25 - 150 + 125 = 0$. ✓

4 CHANGE OF BASE

Solve exactly for x :

$$\log_2(x - 1) + \log_4(x - 1) = 6.$$

Use the change-of-base formula, and state the domain restriction before solving.

SOLUTION *Domain first.* Both logarithms take the input $x - 1$, which must be positive, so $x - 1 > 0$, that is $x > 1$. Everything below is carried out on that domain.

Change of base. Writing $\log_4 t = \frac{\log_2 t}{\log_2 4} = \frac{\log_2 t}{2}$ puts both terms in base 2. With $t = x - 1$ the equation becomes

$$\log_2(x - 1) + \frac{1}{2} \log_2(x - 1) = 6 \iff \frac{3}{2} \log_2(x - 1) = 6 \iff \log_2(x - 1) = 4.$$

Hence $x - 1 = 2^4 = 16$, so $x = 17$, which satisfies $x > 1$.

Check. $\log_2 16 + \log_4 16 = 4 + 2 = 6$. ✓

Part B · Reasoning and modelling

5 LOGARITHMIC EQUATION

Solve over the real numbers:

$$\log_2(x - 1) + \log_2(x + 1) = 3.$$

State the domain before combining the logarithms. Check every possible answer in the original equation.

SOLUTION Both logarithms require positive inputs: $x - 1 > 0$ and $x + 1 > 0$. Therefore $x > 1$. On this domain, the product law gives

$$\log_2((x - 1)(x + 1)) = 3 \iff x^2 - 1 = 8 \iff x^2 = 9.$$

The algebra gives $x = 3$ or $x = -3$, but the domain keeps only $x = 3$. Check: $\log_2(2) + \log_2(4) = 1 + 2 = 3$.

6 GROWTH COMPARISON

Put the following functions in order from slowest growth to fastest growth as the positive integer n becomes very large:

$$\log_2 n, \quad n, \quad n \log_2 n, \quad n^2, \quad 2^n.$$

In two or three sentences, explain why checking one value such as $n = 3$ does not prove the full ranking.

SOLUTION The eventual order is

$$\log_2 n < n < n \log_2 n < n^2 < 2^n.$$

This is an eventual statement: it becomes true once n is large enough, not necessarily at every positive integer. For example, at $n = 3$, $2^n = 8 < 9 = n^2$, but for sufficiently large n the exponential 2^n is larger than n^2 .

7 PERIODIC MODEL

Let t be the number of hours after midnight. A simple model for temperature is

$$T(t) = 3 \sin\left(\frac{\pi}{6}(t - 1)\right) + 20 \quad (0 \leq t \leq 12).$$

- State the amplitude, midline, period, and range.
- At what time does the model first reach a maximum? At what time does it reach a minimum?
- Explain why $(t - 1)$ moves the graph right by 1 hour, rather than left by 1 hour.

SOLUTION The amplitude is **3**, the midline is $y = 20$, the period is $\frac{2\pi}{\pi/6} = 12$ hours, and the range is $[17, 23]$. A maximum occurs when the sine input is $\pi/2$, so $\frac{\pi}{6}(t - 1) = \frac{\pi}{2}$ gives $t = 4$. A minimum occurs when the input is $3\pi/2$, giving $t = 10$. In $f(t - 1)$, the new input must be $t = 1$ to give the old value $f(0)$, so the graph moves right by 1.

Part C · Communicate precisely

8 FIND THE ERROR

A student writes: “ $\log_2(u + v) = \log_2 u + \log_2 v$ because logarithms turn an operation into addition.”

- Identify the false statement.
- Give a counterexample with positive u and v .
- Write the correct logarithm law for a product, including its condition.

SOLUTION The false statement is $\log_2(u + v) = \log_2 u + \log_2 v$. Take $u = v = 1$: $\log_2(1 + 1) = \log_2 2 = 1$, while $\log_2 1 + \log_2 1 = 0 + 0 = 0$. The correct product law is $\log_2(uv) = \log_2 u + \log_2 v$ for $u > 0$ and $v > 0$.

The statement “An exponential function is always bigger than a polynomial” is false or unclear as written. Rewrite it as a true statement about c^n and n^k , including all important conditions. Then give one numerical example showing why the words “for sufficiently large n ” are necessary.

SOLUTION One correct version is: “For every fixed $c > 1$ and $k > 0$, there is a positive integer N such that $c^n > n^k$ for every integer $n \geq N$.” The word “fixed” matters: the bases and powers are not changing with n . For $c = 2$, $k = 2$, and $n = 3$, we have $2^3 = 8 < 9 = 3^2$, so the inequality is not true for every positive n .